

TED UNIVERSITY

FACULTY OF ENGINEERING

DEPARTMENT OF COMPUTER ENGINEERING

CMPE 491 | SENIOR DESIGN PROJECT I

PerDrobe

Project specifications report

Project team

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1. Introduction

1.1 Project description

PerDrobe is the mobile app we are proposing for our CMPE 491-492 senior design project. We started with a common problem: people save outfit ideas online, but the clothes they already have at home do not always come to mind when they get dressed. We want the app to help users keep track of those clothes and use them when making an outfit. The project proposal gives the overall idea and the features we hope to build [1]. We have also followed the CMPE 491 syllabus requirements for this report [2].

A user will be able to photograph a clothing item and add it to a private digital wardrobe. We want the app to remove the background so the item is easier to see alongside other clothes. It will also suggest simple details, such as category and color. The user should be able to fix a wrong label instead of having to accept whatever the image model predicts. Items in the wardrobe can then be arranged into outfits and saved for later.

The recommendation feature is the main technical part of the project. A user might ask what to wear to dinner, for example, and get a few combinations made from their own clothes. They will be able to give style preferences, including different fit preferences for different clothing categories. We also want a discovery page where users choose to share outfits and where others can like or save what they see. These actions may give us useful preference signals, although we will need to test whether they actually help.

Outfit Remix connects the discovery page to the personal wardrobe. When someone chooses Remix on a shared outfit whose pieces are already identified, the system will look for suitable alternatives among the clothes they own. For example, it might substitute a beige jacket with another jacket in their wardrobe, then check whether the combination still works. If a reasonable match is not possible, the app should say so. We would also like to suggest products from stores and offer limited size guidance, but these features depend on reliable product data and size charts [1].

1.1.1 Planned scope

For the capstone prototype, we are putting the digital wardrobe, photo processing, manual outfit creation, personalized recommendations and a basic discovery page first. Outfit Remix is also a major goal, starting with outfits whose pieces are already identified in our system. Understanding every item in an uncontrolled full-body photo will be harder, so that part needs its own experiments. We are not including photo-realistic virtual try-on or an internet-wide search for the exact product shown in a picture.

Our main research question is whether using personal preferences and wardrobe information improves outfit recommendations. We plan to compare three versions: one based on clothing attributes, one that also uses user preferences and interactions, and one that includes the user's wardrobe. These are planned experiments, not results we have already obtained.

1.1.2 Initial literature review

We first looked at work on outfit compatibility. Han et al. [3] used a bidirectional LSTM to learn relationships between pieces in an outfit and to suggest missing items. Vasileva et al. [4] studied representations that take clothing types into account. Their work is useful for Outfit Remix because two garments can look alike without being suitable substitutes in a whole outfit. It also explains why choosing individually similar items may produce a combination that does not work well together.

Our other concern is making recommendations for a particular person. Ding et al. [5] combined preference information with outfit generation. Tangseng et al. [6] studied recommendations built from items in a personal closet, which is close to one of our main use cases. We can use these approaches as

references, but we still need to decide how to represent wardrobe availability and compare it fairly with the other inputs.

The app will also need to understand clothing photographs. Fashionpedia [7] covers clothing categories, segmentation and fine-grained visual attributes. FashionCLIP [8] shows how images and text descriptions of fashion products can be represented for retrieval and related tasks. We expect to use a suitable pretrained visual model for image features, then train a smaller compatibility or ranking model ourselves. The choice of model is still open until we run tests on our available hardware.

Polyvore is one possible source for compatibility experiments [9]. Its repository provides outfit records and splits, but the old image URLs no longer work. We need to check which images we can access and what the usage conditions allow. We will also test with ordinary phone photos because they differ from clean catalog images. At this point, the literature helps us choose experiments; it does not prove that our own implementation will give good recommendations.

1.2 Constraints

1.2.1 Economic constraints

We would like to finish the capstone without paying for AI APIs, software subscriptions or rented GPUs. Our team has computers with RTX 4060-class graphics cards, which we can use for model experiments. Free database and hosting services may be enough for a small test group. Those services have limits, so the prototype should still work in a local development setup. Keeping the app online for outside users at all hours is a possible later step with a separate budget.

1.2.2 Environmental constraints

Image processing and model training use electricity, and storing both original and processed photos takes space. We will try to process a clothing photo when it is added, keep the features we need, and reuse them for later recommendations. Smaller image files and a compact ranking model should also help us avoid unnecessary processing. We cannot say the app has a positive environmental effect without measuring it.

1.2.3 Social constraints

Clothing choices are personal. The app needs to work for users with different styles and body types, including people who do not want to enter measurements. We will try to keep the main screens simple and readable and let users correct automatic labels. The discovery page will also need a way to report inappropriate posts. A large social feed is not necessary for the first version.

1.2.4 Political and legal constraints

Personal photos, optional measurements and interaction records can fall under data protection requirements. We need to handle them in line with applicable Turkish legislation, including Law No. 6698 (KVKK) [12]. The source and permitted use of dataset images, brand photographs and product catalogs must also be checked. We do not expect political concerns to determine the project, but rules about content, privacy and platform use still affect how we design it.

1.2.5 Ethical constraints

The model may favor styles that appear more often in the training data, and its suggestions can be wrong. Users should keep control of their own wardrobe information. We will not present a size suggestion as a guarantee that a product will fit. We also need to be clear about how likes and saved outfits are used for personalization. When we evaluate the model, unsuccessful predictions matter as much as successful examples.

1.2.6 Health and safety constraints

PerDrobe is for clothing suggestions, not health assessments. Body measurements, if provided, will only support optional size guidance. The app should not require body photographs or comment negatively on a user's appearance. We will minimize the personal information requested for each feature.

1.2.7 Manufacturability and technical feasibility

Since this is a software project, manufacturability mainly means being able to build, test and run the full system with our available tools. The mobile app, database, API and image-processing code need clear interfaces. We want to test them together early, beginning with uploading one garment and receiving one recommendation. Keeping the backend and AI code separate from the mobile screens will also make a later move to hosted servers easier.

1.2.8 Sustainability and maintenance

The project should be understandable to someone who needs to continue it after graduation. We will keep the code in version control, document the development setup and record the datasets and model versions used in experiments. User data should have a backup and deletion plan. We hope that recommending combinations from existing clothes may encourage people to use what they own, but we will not claim that this reduces consumption unless we can show evidence.

1.3 Professional and ethical issues

We will use the ACM Code of Ethics [10] and the Software Engineering Code of Ethics [11] as guides for our work. The most immediate responsibility is to protect user data. A private wardrobe should stay private unless its owner deliberately shares an outfit. The backend must check who is allowed to access or change each item. We will keep server credentials out of the mobile app, explain why information is collected and provide a way to delete personal records. Any use of user photos or interactions for research or model training needs its own appropriate permission and legal basis [12].

We also have responsibilities when using material created by other people. We will cite papers and software we rely on, check the licenses of models and datasets, and avoid copying store photographs without permission. If a dataset is available for research but not for a public product, we will keep that restriction recorded. The same applies to the images shown in the discovery feed.

For the AI part, we will report results from separate training and test data and explain where the model fails. Good benchmark scores would not automatically prove that suggestions suit real users, so we would like to include a small user evaluation as well. We will keep track of each team member's contribution and document the changes we make. These are practical checks we can follow during development, not just statements for the final report.

2. Requirements

The requirements below describe the version we are aiming to build. "Core" means we need the feature for the capstone demonstration. "Planned" means we want to add it after the main system works, if time allows. "Conditional" means it needs reliable outside data or permission; it is not required for the wardrobe-to-outfit flow. Where we have chosen an initial limit, such as photo size, we can revise it after testing and document the decision. The next Analysis Report will contain more detailed test cases.

2.1 Functional requirements

A saved outfit stays private until its owner chooses to publish it (FR-19). Remix begins from an eligible public post (FR-20) and searches only the requesting user's wardrobe (FR-12).

ID	Requirement	Priority
FR-01	Users can create an account, sign in, sign out and edit basic profile details.	Core
FR-02	Users can add, view, update and remove their own wardrobe items with clothing photos. Photo uploads must pass the checks in FR-23 before they are processed.	Core
FR-03	When a user uploads a garment photo, the app tries to remove its background. If this fails, the original photo is kept unchanged. The user can retry processing or save the item with the original photo.	Core
FR-04	The app suggests an item category and color. Users can change incorrect information.	Core
FR-05	The system shall allow users to enter and update their style preferences and category-specific clothing fit preferences. Body measurements are not required for core recommendations.	Core
FR-06	Users can arrange their wardrobe items into an outfit and save or edit the result.	Core
FR-07	Users can request outfits made from their own wardrobe for an occasion, such as a dinner or a party. If the wardrobe does not contain enough suitable pieces, the system explains that it cannot make a suitable outfit instead of showing items the user does not own.	Core
FR-08	The system generates candidate outfits and ranks them using clothing features and a learned compatibility score.	Core
FR-09	The recommendation system shall rank outfits using the user's stated style and category-specific fit preferences, even when no interaction data is available.	Core
FR-10	Users can browse public outfit posts in Discovery and open a post to see its outfit. Private wardrobe items and unpublished outfits must not appear there.	Core
FR-11	Users can like and unlike a public outfit post. The app shows the current like state and keeps it when the user opens the post again.	Planned
FR-12	When a user requests Outfit Remix for a public outfit whose garment categories are identified, the system searches only that user's private wardrobe for suitable alternatives. It proposes an outfit using owned items, or clearly says that no suitable combination could be found. This does not require analyzing every garment in an unrestricted full-body photograph.	Core
FR-13	The app shows the items in a suggested outfit together in a simple digital composition.	Core
FR-14	If a product catalog is available and its usage permissions allow it, the system can recommend products that go with the user's wardrobe and link to the products on external retailer websites. Purchases and checkout are handled by the retailer, not PerDrobe.	Conditional
FR-15	If usable product-specific size charts are available and the user requests size guidance, the app can ask for relevant optional body measurements, allow the user to update them, and give a cautious size suggestion. If the information is insufficient, it must say so. No measurements are needed when this conditional feature is not used.	Conditional
FR-16	Users can delete their own items, saved outfits, posts and account-related data through defined controls.	Core
FR-17	Signed-in users can report a public outfit post and provide a reason. The report is saved with a reference to the post. During the capstone prototype, the project team will review reports through restricted backend access, record each decision, and remove a reported post from Discovery if necessary. A separate moderator screen is not required.	Core
FR-18	The system records the model version and basic response-time results needed for evaluation, without logging private images unnecessarily.	Core
FR-19	Users can publish one of their own saved outfits to Discovery only after choosing Share and confirming what will be visible. They can later unpublish it. Only explicitly published outfits become visible to other users; publishing or unpublishing must not expose or delete the rest of the private wardrobe.	Core

ID	Requirement	Priority
FR-20	For an eligible public outfit in Discovery whose garment categories are identified, users can select Remix to start FR-12. If the post lacks the information required for Remix, the option must be unavailable or the app must explain why it cannot proceed.	Core
FR-21	Users can save and unsave a public outfit post in a private Saved list. The list must show the current saved state when reopened.	Planned
FR-22	Users can follow and unfollow other users through their public profiles. The profile must show the current follow state when reopened.	Planned
FR-23	The app shall accept JPEG or PNG wardrobe photos up to an initial 10 MB file-size limit. Before processing, it checks the actual file type and size. Unsupported or oversized files are rejected with a clear reason, without creating or changing a wardrobe item.	Core

2.2 Non-functional requirements

ID	Area	Requirement
NFR-01	Privacy	Wardrobe items are private by default. Only an outfit explicitly published by its owner is shown to others. The backend must check authorization before private data is returned, changed or accessed through a public post.
NFR-02	Security	The app must use authenticated sessions and encrypted remote API connections. Passwords must never be stored as plain text in the application database; credential handling shall be delegated to a trusted authentication service. Privileged keys remain on the server. Requests without valid authorization must not access private records.
NFR-03	Error handling	Unsupported uploads are rejected before processing without changing saved wardrobe items (FR-23). If background removal fails on a valid upload, the original photo remains available for retry or saving without processing (FR-03). Recommendation errors show a clear message and leave existing wardrobe items unchanged.
NFR-04	Usability	A usability test shall include at least five first-time users. After a short introduction, each participant will try three tasks without step-by-step help: adding a garment, saving an outfit, and requesting a recommendation. For each task, at least 80% of all participants shall finish independently. We will record task completion, errors, requests for help, and where users get stuck.
NFR-05	Response time	On an Android test phone connected to the same Wi-Fi as an RTX 4060-class backend, we will time (1) uploading and processing a 1024 x 1024 pixel garment photo until it appears in the wardrobe and (2) requesting an outfit from a fixed 50-item wardrobe until suggestions appear. With the model already loaded, at least 9 of 10 repeated requests shall finish within 20 seconds for processing and 5 seconds for recommendations. We will also report cold starts and results with 200 wardrobe items, and record the phone, GPU, connection and model version.
NFR-06	Portability	The mobile application shall support both locally deployed and remotely hosted backends without requiring changes to the API request and response structures. We will test this by connecting the same app build to both environments using only configuration changes.
NFR-07	Maintainability	Each major system module shall have documented setup instructions and clearly defined interfaces for communicating with other modules. The documentation will describe the inputs, outputs, and dependencies of the wardrobe, discovery, image-processing, and recommendation modules.
NFR-08	Data consistency	Photos, garments, outfits and posts stay linked to their correct owners. Deleted private content must not remain publicly accessible.
NFR-09	Accessibility	We will check the main screens with a contrast checker, text-size tests and Android accessibility tools. Text should meet WCAG 2.2 AA contrast guidance: at least 4.5:1 for normal text and 3:1 for large text [13]. Main controls should have labels that a screen reader can announce. Enlarging system text must not hide essential content or controls.

ID	Area	Requirement
NFR-10	Honest output	Recommendations are shown as suggestions. The app should not promise an exact fit or a guaranteed matching outfit.
NFR-11	Cost	A full capstone demo must be possible on existing team hardware without requiring a paid AI API.
NFR-12	Reproducibility	We will record data splits, model versions and experiment settings for any reported AI comparison.

These usability and timing limits are initial targets for our prototype, not results we have already achieved. We will check whether they are realistic in an early pilot, discuss any changes with our supervisor and fix the acceptance criteria before the final tests. The test record will show each condition, the expected result, the measured result and whether the requirement passed.

2.3 Data and AI evaluation

We intend to use suitable existing models for image-related tasks and train a smaller outfit compatibility or recommendation model ourselves. We have not chosen the final training dataset yet. Before doing that, we need to check whether the images and labels are accessible and whether the license allows our intended use. Clothing information suggested by the AI will remain editable by the user.

Our comparison will use the three configurations described in Section 1.1.1. We will control the candidate outfits and testing conditions where possible, so one method does not gain an advantage simply by having different options. With suitable relevance labels, we can report Precision@K or NDCG@K. We also want to measure diversity, wardrobe coverage and response time. A small user study would help us understand whether people actually prefer the outfits. Training and test items must be kept separate, and near-duplicates need special care [3, 4, 9].

Some results may be difficult to interpret. Replacing an item in a known outfit gives us a comparison example, but the replacement is not necessarily a bad fashion choice. We will record those limitations instead of treating every generated negative example as a definite mistake. Size guidance will only be assessed if we get usable product-specific reference data.

2.4 First implementation target

Our first end-to-end test will be small: upload a garment photo, process it, save it in the user's wardrobe and return one basic outfit suggestion. Then we can work on better ranking, discovery interactions and Outfit Remix. For the capstone we expect to run some services on our own computers. We are keeping the system modular so that moving it to a continuously available server later would not require rebuilding the mobile app. This report lists our initial requirements, not completed features.

3. References

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